Divided Differences and Newton's Divided Difference Interpolation

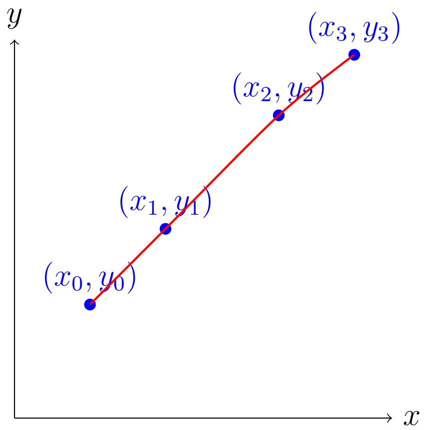
Teacher-Style Notes

1 Introduction

Interpolation is the process of estimating values of a function at points within the range of known data points. It allows us to approximate a continuous curve when only discrete points are known.

Divided differences provide a systematic way to construct interpolating polynomials incrementally, and are particularly useful for Newton's interpolation method.

Diagram of Data Points



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2 Divided Differences

2.1 Definition

Given $n+1$ data points $(x_0, y_0), \ldots, (x_n, y_n)$:

- **Zero-order:** $f[x_i] = y_i$
- **First-order:**

$$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i}$$

- **Higher-order:** recursively,

$$f[x_i, x_{i+1}, \ldots, x_{i+k}] = \frac{f[x_{i+1}, \ldots, x_{i+k}] - f[x_i, \ldots, x_{i+k-1}]}{x_{i+k} - x_i}$$

Intuition: Each higher-order divided difference measures the change of the previous order differences, capturing slope, curvature, and higher-level variations.

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3 Newton's Divided Difference Formula

The Newton interpolating polynomial is:

$$P_n(x) = f[x_0]+f[x_0, x_1](x-x_0)+f[x_0, x_1, x_2](x-x_0)(x-x_1)+\cdots+f[x_0, \ldots, x_n](x-x_0)\ldots(x-x_{n-1})$$

Explanation of terms:

- $f[x_0]$ = value at first point
- $f[x_0, x_1](x - x_0)$ = linear term
- $f[x_0, x_1, x_2](x - x_0)(x - x_1)$ = quadratic term
- Higher-order terms follow similarly

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4 Simplified Notation

The polynomial can be compactly written as:

$$P_n(x) = \sum_{k=0}^{n} f[x_0, x_1, \ldots, x_k] \prod_{j=0}^{k-1} (x - x_j)$$

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5 Error Analysis

The interpolation error at a point x is:

$$R_n(x) = f(x) - P_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n} (x - x_i), \quad \xi \in [\min(x_i), \max(x_i)]$$

Intuition: The error depends on the next derivative of f and the product of differences from all known points.

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6 Worked Example

Data Points

$$(x_0, y_0) = (1, 2), \quad (x_1, y_1) = (2, 3), \quad (x_2, y_2) = (4, 7)$$

Step 1: Compute Divided Differences

x_i	$f[x_i]$	1st	2nd
1	2	$f[1,2] = (3-2)/(2-1) = 1$	$f[1,2,4] = (2-1)/(4-1) = 1/3$
2	3	$f[2,4] = (7-3)/(4-2) = 2$	
4	7		

Step 2: Construct Newton Polynomial

$$\begin{aligned} P_2(x) &= f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) \\ &= 2 + 1(x-1) + \frac{1}{3}(x-1)(x-2) \\ &= \frac{1}{3}x^2 - x + 3 \end{aligned}$$

Verification:

$$P_2(1) = 2, \quad P_2(2) = 3, \quad P_2(4) = 7$$

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7 Summary Table

Concept	**Formula / Notes**
Zero-order divided difference	$f[x_i] = y_i$
First-order divided difference	$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i}$
Higher-order divided differences	$f[x_i, ..., x_{i+k}] = \frac{f[x_{i+1}, ..., x_{i+k}] - f[x_i, ..., x_{i+k-1}]}{x_{i+k} - x_i}$
Newton interpolating polynomial	$P_n(x) = \sum_{k=0}^{n} f[x_0, ..., x_k] \prod_{j=0}^{k-1} (x - x_j)$
Error formula	$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n} (x - x_i)$
Advantages	Incremental, flexible spacing, easy evaluation
Limitations	Oscillations for high-degree, sensitive to noise, overfitting